a) Write down the population model you will use to address this question (recall: the outcome variable has 5 categories, so you should include four logit equations). Be sure your notation is clear. (8 points)

log **P {(**Y =mental health**)/ P (**Y = medical**)}** = $\beta_{omh} + \beta_{1mh}*X_{raceth2} + \beta_{2mh}*X_{raceth3} + \beta_{3mh}*X_{raceth4} + \beta_{4mh}*X_{raceth5} + \beta_{5mh}*X_{bor1} + \beta_{6mh}*X_{bor3} + \beta_{7mh}*X_{bor4} + \beta_{8mh}*X_{bor5} + \beta_{9mh}*X_{bor6} + \beta_{10mh}*X_{preg2} + \beta_{11mh}*X_{preg3}$

log **P {(**Y =homicide**)/ P (**Y = medical**)}**= $\beta_{oh} + \beta_{1h}*X_{raceth2} + \beta_{2h}*X_{raceth3} + \beta_{3h}*X_{raceth4} + \beta_{4h}*X_{raceth5} + \beta_{5h}*X_{bor1} + \beta_{6h}*X_{bor3} + \beta_{7h}*X_{bor4} + \beta_{8h}*X_{bor5} + \beta_{9h}*X_{bor6} + \beta_{10h}*X_{preg2} + \beta_{11h}*X_{preg3}$

log **P {(**Y =unintentional injuries**)/ P (**Y = medical**)}** = $\beta_{oui} + \beta_{1ui}*X_{raceth2} + \beta_{2ui}*X_{raceth3} + \beta_{3ui}*X_{raceth4} + \beta_{4ui}*X_{raceth5} + \beta_{5ui}*X_{bor1} + \beta_{6ui}*X_{bor3} + \beta_{7ui}*X_{bor4} + \beta_{8ui}*X_{bor5} + \beta_{9ui}*X_{bor6} + \beta_{10ui}*X_{preg2} + \beta_{11ui}*X_{preg3}$

log **P {(**Y =unknown/other**)/ P (**Y = medical**)}**= $\beta_{ouo} + \beta_{1uo}*X_{raceth2} + \beta_{2uo}*X_{raceth3} + \beta_{3uo}*X_{raceth4} + \beta_{4uo}*X_{raceth5} + \beta_{5uo}*X_{bor1} + \beta_{6uo}*X_{bor3} + \beta_{7uo}*X_{bor4} + \beta_{8uo}*X_{bor5} + \beta_{9uo}*X_{bor6} + \beta_{10uo}*X_{preg2} + \beta_{11uo}*X_{preg3}$

Where, $X_{raceth2}$: 1 = Hispanic, 0 = otherwise, $X_{raceth3}$: 1 = Asian/Pacific Islanders, 0 = otherwise $X_{raceth4}$: 1 = Non-Hispanic White, 0 = otherwise $X_{raceth5}$: 1 = Other race, 0 = otherwise

X_{bor1} : 1 = Brooklyn, 0 = otherwise X_{bor3} : 1 = Manhattan, 0 = otherwise X_{bor4} : 1 = Queens, 0 = otherwise X_{bor5} : 1 = Staten Island, 0 = otherwise X_{bor6} : 1 = Rest of the State, 0 = otherwise

X_{preg2} : 1 = Pregnancy associated, 0 = otherwise X_{preg3} : 1 = Unable to determine, 0 = otherwise

b) Using R, conduct a formal test to determine whether race/ethnicity is significantly associated with cause of death, controlling for borough of residence and whether death is related to pregnancy. To respond: write down the null

and alternative hypotheses, test statistic with appropriate degrees of freedom, p-value, conclusion. Use the 5% level of significance. Include the appropriate output in the answer. (15 points)

b. Test to see the validity of the race for the association of cause of death controlling for borough and death related pregnancy.

```
Likelihood ratio test

Model 1: cause2 ~ raceth + bor + preg
Model 2: cause2 ~ bor + preg
  #Df  LogLik  Df  Chisq Pr(>Chisq)
1  48 -249.97
2  32 -260.32 -16 20.709     0.1899
> |
```

H_0: Reduced model is sufficient i.e. race/ethnicity is not significantly associated with cause of death, controlling for borough of residence and whether death is related to pregnancy

H_A: Full model is better than reduced model i.e that race/ethnicity is significantly associated with cause of death, controlling forborough of residence and whether death is related to pregnancy

Test statistic, Chi2 df=16 = 20.709 With df=16, $p\ value$ = 0.1899 which is greater than α, 0.05

So, we fail to reject the H_0.

Conclusion:

At the 5% level of significance, the data does not provide enough evidence to conclude that race/ethnicity is significantly associated with cause of death, controlling for borough of residence and whether death is related to pregnancy.

a. Interpretation of the appropriate odds ratios and 95% confidence intervals associated with race/ethnicity

I am interpreting only the significant Odds Ratios (ORs) and their corresponding confidence intervals associated with race/ethnicity.

```
3/raceth5      1.9318e-07 1.9307e-07     0.0000 < 2.2e-16 ***
```

The odds of having homicide versus medical (ref.) as underlying cause of death is 99.999980682% lower (almost certain not to occur) for other race compared to non-Hispanic black, holding borough of residence and death related to pregnancy constant.

We are 95% confident that the true odds ratio for homicide versus medical as underlying cause of death for other race compared to non-Hispanic black could be as low as 0.00 or as high as 0.00000019307. This indicates statistically significant relationship between race and having homicide versus medical as underlying cause of death because the confidence interval does not include the null value OR = 1.

```
4/raceth3      2.6858e+01 1.2904e+00  559.0087 0.0336234 *
```

The odds of having unintentional injuries versus medical (ref.) as underlying cause of death is 2585.8% higher for Asian/Pacific Islanders compared to non-Hispanic black, holding borough of residence and death related to pregnancy constant.

We are 95% confident that the true odds ratio for unintentional injuries versus medical as underlying cause of death for Asian/Pacific Islanders compared to non-Hispanic black could be as low as 1.2904 or as high as 559.0087. This indicates statistically significant relationship between race and having unintentional injuries versus medical as underlying cause of death because the confidence interval does not include the null value OR = 1.

```
> questionr::odds.ratio(model)
                    OR      2.5 %    97.5 %          p
1/(Intercept) 3.0488e-01 1.3640e-01    0.6815 0.0037968 **
1/raceth2     5.4645e-01 2.1287e-01    1.4028 0.2089998
1/raceth3     1.4484e-01 1.7671e-02    1.1872 0.0718445 .
1/raceth4     1.1955e+00 4.9090e-01    2.9116 0.6941359
1/raceth5     4.5490e-01 7.8678e-02    2.6302 0.3789660
1/bor1        8.7187e-01 3.2654e-01    2.3279 0.7843610
1/bor3        4.7794e-01 8.4276e-02    2.7104 0.4043800
1/bor4        1.0963e+00 4.1399e-01    2.9029 0.8532435
1/bor5        3.2639e-01 2.8034e-02    3.8002 0.3713268
1/bor6        1.4170e+00 4.7320e-01    4.2432 0.5333819
1/preg2       1.3788e+00 6.2319e-01    3.0505 0.4279189
1/preg3       6.2311e+00 2.0591e+00   18.8561 0.0012018 **
3/(Intercept) 1.0978e-02 1.0118e-03    0.1191 0.0002080 ***
3/raceth2     1.1535e+00 2.3589e-01    5.6405 0.8600272
3/raceth3     1.5978e+00 2.4093e-01   10.5967 0.6273128
3/raceth4     4.4482e-01 6.3152e-02    3.1332 0.4160258
3/raceth5     1.9318e-07 1.9307e-07    0.0000 < 2.2e-16 ***
3/bor1        5.9684e-01 7.9353e-02    4.4890 0.6161362
3/bor3        1.0483e-07 1.0483e-07    0.0000 < 2.2e-16 ***
3/bor4        1.4000e+00 2.5479e-01    7.6931 0.6986953
3/bor5        3.1564e+00 4.1082e-01   24.2508 0.2692242
3/bor6        2.3562e+00 3.2981e-01   16.8324 0.3929428
3/preg2       8.5912e+00 9.0424e-01   81.6247 0.0611607 .
3/preg3       9.9332e+01 9.5710e+00 1030.9136 0.0001171 ***
4/(Intercept) 7.4434e-04 1.3404e-05    0.0413 0.0004406 ***
4/raceth2     2.3183e+00 1.1954e-01   44.9601 0.5783222
4/raceth3     2.6858e+01 1.2904e+00  559.0087 0.0336234 *
4/raceth4     6.3161e+00 4.5082e-01   88.4897 0.1711727
4/raceth5     2.7019e-06 0.0000e+00       Inf 0.9901759
4/bor1        5.4326e-12 5.4326e-12    0.0000 < 2.2e-16 ***
4/bor3        1.9906e-06 1.9872e-06    0.0000 < 2.2e-16 ***
4/bor4        9.4406e+00 8.5545e-01  104.1854 0.0668732 .
4/bor5        1.7043e-07 1.7042e-07    0.0000 < 2.2e-16 ***
4/bor6        5.2153e+00 2.3801e-01  114.2784 0.2943623
4/preg2       7.4598e+00 5.3771e-01  103.4908 0.1342401
4/preg3       9.1089e+01 3.9669e+00 2091.5752 0.0047757 **
5/(Intercept) 2.3925e-01 1.0407e-01    0.5500 0.0007585 ***
5/raceth2     1.2791e+00 5.2866e-01    3.0947 0.5850730
5/raceth3     8.2625e-01 2.2785e-01    2.9962 0.7715202
5/raceth4     1.0058e+00 3.5394e-01    2.8581 0.9913618
5/raceth5     7.2094e-01 1.2666e-01    4.1037 0.7123127
5/bor1        1.4429e+00 5.8795e-01    3.5409 0.4234594
5/bor3        8.3613e-01 1.7741e-01    3.9407 0.8209966
5/bor4        9.6239e-01 3.3790e-01    2.7410 0.9427646
5/bor5        1.8496e-07 1.8495e-07    0.0000 < 2.2e-16 ***
5/bor6        6.0734e-01 1.5494e-01    2.3807 0.4743201
5/preg2       1.1253e+00 5.0413e-01    2.5120 0.7732004
5/preg3       4.1332e+00 1.3134e+00   13.0065 0.0152613 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
+ >
```

d.

(i). Probability of death from mental health condition for a pregnancy-related maternal death of a resident of Manhattan identifying as Asian or Pacific Islander

```
> # Asian or pacific islander
> OR_data = data.frame(raceth = "3", bor = "3", preg = "1")
> OR_Probable <- predict(model, OR_data, type = "probs")
> OR_Probable
            2             1             3             4             5
8.428924e-01 1.778928e-02 1.549953e-09 3.354331e-08 1.393182e-01
```

The probability of death from mental health condition for a pregnancy-related maternal death of a resident of Manhattan identifying as Asian or Pacific Islander is approximately 0.0178

(ii). Probability of death from mental health condition for a pregnancy-related maternal death of a resident of Manhattan identifying as non-Hispanic Black

```
> # Non-hispanic black
> OR_data = data.frame(raceth = "1", bor = "3", preg = "1")
> OR_Probable <- predict(model, OR_data, type = "probs")
> OR_Probable
            2             1             3             4             5
7.430754e-01 1.082766e-01 8.551594e-10 1.101034e-09 1.486480e-01
```

The probability of death from mental health condition for a pregnancy-related maternal death of a resident of Manhattan identifying as non-Hispanic black is approximately 0.1083

(iii). Probability of death from mental health condition for a pregnancy-related maternal death of a resident of Manhattan identifying as Hispanic

```
> # Hispanic
> OR_data = data.frame(raceth = "2", bor = "3", preg = "1")
> OR_Probable <- predict(model, OR_data, type = "probs")
> OR_Probable
            2             1             3             4             5
7.487857e-01 5.962240e-02 9.940048e-10 2.572163e-09 1.915919e-01
```

The probability of death from mental health condition for a pregnancy-related maternal death of a resident of Manhattan identifying as Hispanic is approximately 0.0596

(iv). Probability of death from mental health condition for a pregnancy-related maternal death of a resident of Manhattan identifying as non-Hispanic white

```
> # Non-Hispanic White
> OR_data = data.frame(raceth = "4", bor = "3", preg = "1")
> OR_Probable <- predict(model, OR_data, type = "probs")
> OR_Probable
           2            1            3            4            5
7.270571e-01 1.266578e-01 3.721936e-10 6.804305e-09 1.462851e-01
```

The probability of death from mental health condition for a pregnancy-related maternal death of a resident of Manhattan identifying as non-Hispanic white is approximately 0.1267

Appendix: R codes used for this assignment

```
rm(list=ls())
setwd("C:/Users/HP PAVILLION 17/OneDrive/Desktop/Multivariate")
library(readr)
pregmortality <- read.csv("pregmortality.csv")
install.packages("lmtest")
install.packages("car")
install.packages("MASS")
install.packages("lme4")
install.packages("glmnet")
library(questionr)
library(nnet)
library(lmtest)

## Factor the variable for all relevant categories
pregmortality$cause2 <- factor(pregmortality$cause2)
pregmortality$raceth <- factor(pregmortality$raceth)
pregmortality$bor <- factor(pregmortality$bor)
pregmortality$preg <- factor(pregmortality$preg)

## Specify reference categories as given in the questions of the Homework
pregmortality$cause2 <- relevel(pregmortality$cause2, ref = 2)
pregmortality$raceth <- relevel(pregmortality$raceth, ref = 1)
pregmortality$bor <- relevel(pregmortality$bor, ref = 2)
pregmortality$preg <- relevel(pregmortality$preg, ref = 1)
```

```
25
26  ## Fit multinomial logit and later testing it for wald tests
27
28  model <- multinom(cause2 ~ raceth + bor + preg, data = pregmortality)
29  summary(model, Wald.ratios = TRUE)
30
31  ## Partial likelihood to test: race with respect to its relevance in the model.
32  reduced_model <- multinom(cause2 ~ bor + preg, data = pregmortality)
33  lmtest::lrtest(model, reduced_model)
34
35  ##Finding all odds ratios and 95% CI for each questionr::odds.ratio(model)
36  ## Compute predicted probability
37
38  # Asian or pacific islander
39  OR_data = data.frame(raceth = "3", bor = "3", preg = "1")
40  OR_Probable <- predict(model, OR_data, type = "probs")
41  OR_Probable
42
43
44  # Non-hispanic black
45  OR_data = data.frame(raceth = "1", bor = "3", preg = "1")
46  OR_Probable <- predict(model, OR_data, type = "probs")
47  OR_Probable
48
49  # Hispanic
50  OR_data = data.frame(raceth = "2", bor = "3", preg = "1")
51  OR_Probable <- predict(model, OR_data, type = "probs")
52  OR_Probable
53
54  # Non-Hispanic White
55  OR_data = data.frame(raceth = "4", bor = "3", preg = "1")
56  OR_Probable <- predict(model, OR_data, type = "probs")
57  OR_Probable
58
59  # Non-hispanic black
60  OR_data = data.frame(raceth = "4", bor = "3", preg = "1")
61  OR_Probable <- predict(model, OR_data, type = "probs")
62  OR_Probable
63
64
65
```